

CAT-2:Answer Key (Slot-B2)

1. Develop an 8051 Assembly Language Program (ALP) that reads data from Port P1 and performs the following:

- If the most significant bit (P1.7) is 1, perform a bitwise AND of data read from port P1 with the value at Port P2 and store the result at Port P3.
- If the most significant bit (P1.7) is 0, perform a bitwise OR of data read from port P1 with the value at Port P2 and store the result at Port P3.

Solution:

```

liter1  ORG 0000H           ; Start of the program
liter2  MAIN:
liter3  MOV P1, #0FFH      ; Configure P1 as input
liter4  MOV P2, #0FFH      ; Configure P2 as input
liter5  MOV P3, #00H       ; Configure P3 as output
liter6
liter7  READ_LOOP:
liter8  MOV A, P1           ; Read data from Port P1
liter9  JB ACC.7, AND_OP    ; If P1.7 = 1, jump to AND operation
liter10 ; If P1.7 = 0, perform OR operation
liter11 ; =====
liter12 MOV A, P1           ; Load data from P1 again
liter13 ORL A, P2           ; Perform bitwise OR with P2
liter14 MOV P3, A          ; Store result in P3
liter15 SJMP READ_LOOP     ; Repeat loop
liter16
liter17 ; If P1.7 = 1, perform AND operation
liter18 ; =====
liter19 AND_OP:
liter20 MOV A, P1           ; Load data from P1 again
liter21 ANL A, P2           ; Perform bitwise AND with P2
liter22 MOV P3, A          ; Store result in P3
liter23 SJMP READ_LOOP     ; Repeat loop
liter24 END
liter25 ; End of program

```

2. Develop an 8051 ALP to control a camera module connected to Port P1. The camera should take a picture at regular intervals of 1 second. Use Timer 0 in Mode 1 to generate an interrupt every 1 second.

Preconditions:

- It is assumed that the camera only needs a trigger pulse for activation, and therefore a single pin is enough (e.g., P1.0) to take a picture at regular intervals of 1 second.
- XTAL=12MHz to simplify calculation

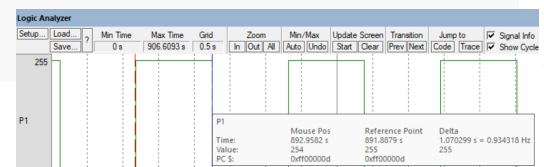
Solution:

- Timer-0 overflows every 65.536ms → 16 overflows give 1.048 sec, which is close to 1 sec.

```

liter1  ORG 0000H
liter2  LJMP MAIN          ; Jump to main program
liter3
liter4  ORG 000BH          ; Timer-0 Interrupt Vector
liter5  TIMERO_ISR:
liter6  DJNZ R2, EXIT_ISR  ; Decrement counter, if not zero, exit
liter7  CPL P1.0           ; Trigger camera connected to P1.0 after 16 overflows (~1 sec
liter8  )
liter9  MOV R2, #16        ; Reset counter for 1-sec delay
liter10 EXIT_ISR:
liter11 MOV TH0, #00H      ; Reload Timer-0 for max delay (65.536ms)
liter12 MOV TLO, #00H
liter13 RETI              ; Return from interrupt
liter14 ; =====

```



```

15 ; Main Program
16 ORG 0030H
17 MAIN:
18 MOV TMOD, #01H ; Timer-0 Mode 1 (16-bit timer)
19 MOV IE, #82H ; Enable Timer-0 interrupt (EA=1, ET0=1)
20 MOV TH0, #00H ; Load Timer-0 for max delay (65.536ms)
21 MOV TLO, #00H
22 SETB TR0 ; Start Timer-0
23 SJMP $ ; Infinite loop to keep program running
24 END

```

3. Develop an 8051 ALP that simulates a simple voice command control system. When a specific serial command (e.g., "LED ON" or "LED OFF") is received through the serial port, the program should turn on/off an LED connected to Port P1.0. The program should check for the voice command every 0.5 second and react accordingly. Use timer 0 to generate the delay and baudrate of 4800.

Algorithm: To check the reception of data every 0.5 seconds, we need to:

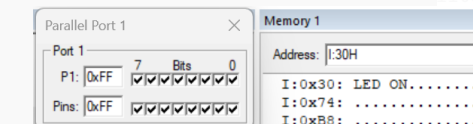
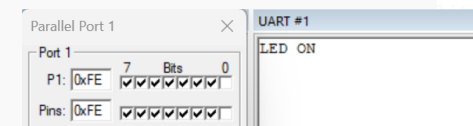
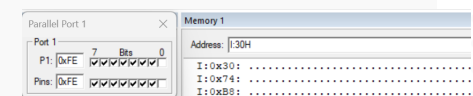
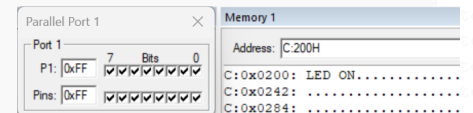
- Configure Timer 0 in Mode 1 (16-bit) to generate a delay of 500ms, and Timer 1 in Mode 2 (for UART).
- Check for received data within this interval.
- If data is received (assumed that data is located in ROM), proceed with comparison.
- If no data is received within 0.5s, restart the loop.

Solution:

```

1 ORG 0000H
2 MOV TMOD, #21H ; Timer 0 in Mode 1 (16-bit), Timer 1 in Mode 2 (for UART)
3 MOV TH1, #0F3H ; Set baud rate 4800 bps, XTAL=12MHz
4 MOV SCON, #50H ; 8-bit data, 1 stop bit, REN enabled
5 SETB TR1 ; Start Timer 1 for UART
6 CLR P1.0 ; Ensure LED is OFF at startup
7
8 AGAIN:
9 MOV R0, #07 ; Counter for 7 characters (including space)
10 MOV R1, #30H ; Store received characters starting at RAM location 30H
11
12 RECEIVE:
13 ACALL DELAY ; Wait for 0.5s before checking
14 JNB RI, RECEIVE ; Wait for received character
15 CLR RI ; Clear receive interrupt flag
16 MOV A, SBUF ; Read received character
17 MOV @R1, A ; Store character in RAM
18 INC R1 ; Move to next RAM location
19 DJNZ R0, RECEIVE ; Repeat until all 7 characters are received
20
21 IGNORE_EXTRA:
22 JNB RI, IGNORE_EXTRA ; Wait for any extra characters
23 CLR RI ; Clear receive interrupt flag
24 SJMP IGNORE_EXTRA ; Keep ignoring extra characters until program resets
25
26 MOV R1, #30H ; Reset RAM pointer after receiving 7 characters
27 MOV DPTR, #200H ; Point to stored "LED ON"
28 MOV R0, #07 ; Counter for 7 characters
29
30 COMPARE_LOOP:
31 CLR A
32 MOVC A, @A+DPTR ; Load reference character from ROM
33 MOV R6, A ; Store reference character in R6
34 MOV A, @R1 ; Load received character from RAM
35 SUBB A, R6 ; Compare received character with reference
36 JNZ NO_MATCH ; If mismatch, exit
37 INC R1 ; Move to next received character
38 INC DPTR ; Move to next reference character
39 DJNZ R0, COMPARE_LOOP ; Repeat for all 7 characters
40
41 SETB P1.0 ; If matched, turn ON LED and hold
42 WAIT_HERE: SJMP WAIT_HERE ; Stay here to keep LED ON
43

```



```

44 NO_MATCH:
45     CLR P1.0          ; Turn OFF LED if mismatch
46     SJMP AGAIN        ; Retry receiving
47
48 ;=====
49 ;**0.5-Second Delay Subroutine (12 MHz Crystal)**
50 DELAY:
51     MOV R2, #08        ; Repeat 8 times (8 * 65.535ms = 524ms)
52 REPEAT_DELAY:
53     MOV TH0, #00H       ; Set Timer 0 High Byte
54     MOV TLO, #00H       ; Set Timer 0 Low Byte
55     SETB TRO            ; Start Timer 0
56
57 WAIT_TIMER:
58     JNB TFO, WAIT_TIMER ; Wait until Timer 0 overflows
59     CLR TRO             ; Stop Timer 0
60     CLR TFO             ; Clear Timer 0 Overflow Flag
61     DJNZ R2, REPEAT_DELAY ; Repeat 8 times
62     RET
63
64     ORG 0200H
65 MESS: DB "LED ON"       ; Stored expected string
66     END

```

4. Develop an 8051 ALP to interface a 16x2 LCD display and periodically update the message shown on the LCD. Use Timer 0 to update the message every 0.5 seconds. Store two strings in ROM: "VIT Campus" at 200H and "VIT University" at 300H. A switch connected to P0.4 toggles between displaying the first string and the second string.

Algorithm:

- Uses Timer 0 to generate a 0.5-second delay.
- Reads P0.4 to toggle between the two strings.
- Updates the LCD dynamically when the switch is pressed.
- Initializes the LCD properly for stable operation.

Solution:

```

1     ORG 0000H
2     LCALL WAIT
3     MOV A, #38H        ; 8-bit mode, 2 lines, 5x7 dots
4     LCALL COMMAND
5     MOV A, #0EH        ; Display ON, cursor ON
6     LCALL COMMAND
7     MOV A, #06H        ; Auto-increment cursor
8     LCALL COMMAND
9     MOV A, #80H        ; Set cursor to line 1, position 1
10    LCALL COMMAND
11    MOV A, #01H        ; Clear display
12    LCALL COMMAND
13
14 MAIN_LOOP:
15     JB P0.4, DISPLAY_UNI ; If switch (P0.4) is HIGH, display "VIT University"
16     LCALL DISPLAY_CAMPUS ; Else, display "VIT Campus"
17     SJMP MAIN_LOOP      ; Repeat checking switch status
18
19 DISPLAY_UNI:
20     LCALL DISPLAY_UNIVERSITY
21     SJMP MAIN_LOOP      ; Return to checking the switch
22
23 ;=====
24 ; Subroutine to display "VIT Campus"
25 DISPLAY_CAMPUS:
26     MOV DPTR, #200H    ; Load ROM pointer for "VIT Campus"
27     MOV RO, #10         ; Number of characters (including space)
28     LCALL DISPLAY_STRING
29     RET
30
31 ;=====

```

```

lite 32 ; Subroutine to display "VIT University"
lite 33 DISPLAY_UNIVERSITY:
lite 34     MOV DPTR, #300H    ; Load ROM pointer for "VIT University"
lite 35     MOV R0, #14       ; Number of characters
lite 36     LCALL DISPLAY_STRING
lite 37     RET
lite 38
lite 39 ;=====
lite 40 ; Subroutine to display a string from ROM to LCD
lite 41 DISPLAY_STRING:
lite 42     MOV R1, #0         ; Internal pointer
lite 43 DISPLAY_LOOP:
lite 44     CLR A
lite 45     MOV A, R1
lite 46     MOVC A, @A+DPTR    ; Get character from ROM
lite 47     MOV B, A
lite 48     LCALL DISPLAY     ; Send character to LCD
lite 49     LCALL DELAY      ; 0.5s delay between character updates
lite 50     INC R1
lite 51     DJNZ R0, DISPLAY_LOOP
lite 52     RET
lite 53
lite 54 ;=====
lite 55 ; Subroutine to send commands to LCD
lite 56 COMMAND:
lite 57     MOV P1, A           ; Send command to Port 1
lite 58     CLR P2.0           ; RS = 0 (Command Mode)
lite 59     CLR P2.1           ; RW = 0 (Write)
lite 60     SETB P2.2          ; E = 1 (Enable)
lite 61     LCALL WAIT        ; Short delay
lite 62     CLR P2.2          ; E = 0 (Latching)
lite 63     LCALL WAIT        ; Short delay
lite 64     RET
lite 65
lite 66 ;=====
lite 67 ; Subroutine to send data to LCD
lite 68 DISPLAY:
lite 69     MOV P1, B           ; Send data to Port 1
lite 70     SETB P2.0          ; RS = 1 (Data Mode)
lite 71     CLR P2.1           ; RW = 0 (Write)
lite 72     SETB P2.2          ; E = 1 (Enable)
lite 73     LCALL WAIT        ; Short delay
lite 74     CLR P2.2          ; E = 0 (Latching)
lite 75     LCALL WAIT        ; Short delay
lite 76     RET
lite 77
lite 78 ;=====
lite 79 ; Subroutine for 0.5-second delay using Timer 0
lite 80 DELAY:
lite 81     MOV TMOD, #01H     ; Timer 0 Mode 1 (16-bit)
lite 82     MOV R2, #08        ; Repeat 8 times (for 0.5 sec)
lite 83 REPEAT_DELAY:
lite 84     MOV TH0, #3CH       ; Set Timer High byte (for 65.535ms delay)
lite 85     MOV TL0, #0B0H     ; Set Timer Low byte
lite 86     SETB TR0          ; Start Timer 0
lite 87
lite 88 WAIT_TIMER:
lite 89     JNB TFO, WAIT_TIMER ; Wait for Timer 0 to overflow
lite 90     CLR TR0            ; Stop Timer
lite 91     CLR TFO            ; Clear Timer Flag
lite 92     DJNZ R2, REPEAT_DELAY ; Repeat 8 times
lite 93     RET
lite 94
lite 95 ;=====
lite 96 ; Short Wait Subroutine for LCD Commands
lite 97 WAIT:
lite 98     MOV R5, #250
lite 99 WAIT_LOOP:
lite 100    DJNZ R5, WAIT_LOOP
lite 101    RET
lite 102
lite 103 ;=====
lite 104 ; Stored messages at ROM locations

```

```

105  ORG 200H
106  MESS1: DB "VIT Campus"      ; First message
107  ORG 300H
108  MESS2: DB "VIT University"  ; Second message
109  END

```

5. An 8051 microcontroller is interfaced with an ADC0804, which is connected to a water level sensor to monitor and measure the water level of a tank. The sensor provides a 5V output for the full tank level (assumed to be at 100). Develop an ALP with suitable interfacing diagram to interface ADC0804 with 8051 and compare the water level with the threshold value of 75. If the water level exceeds 75, activate the buzzer connected to Port P1.5 for 150 ms using Timer 1 to generate the delay.

Algorithm:

- Compares ADC value with the 75% threshold (assuming $5V = 255$ ADC value, 75% 191).
- Activates the buzzer (P3.5) for 150ms when water exceeds 75%.
- Ensures ADC conversion stability with optimized delays.
- Uses Timer 1 for precise timing.

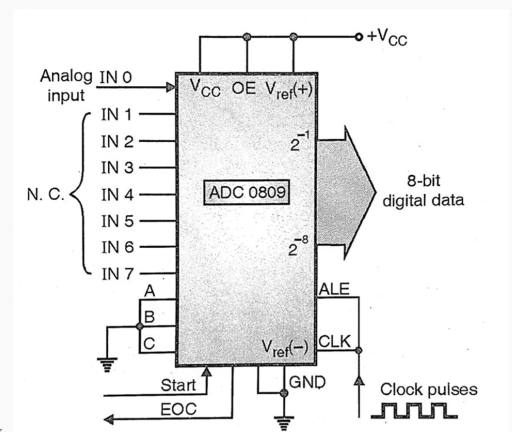
Note: This code uses the ADC0808, but the actual question was for ADC0804. In this code, I assumed that the buzzer is connected to P3.0. as I utilized port 1 for input connection with 8051.

Solution:

```

1  ORG 0000H
2  MOV P1, #0FFH      ; Configure P1 as input (ADC Data)
3  MOV P2, #00H       ; Configure P2 as output (ADC Channel Select)
4  MOV P3, #00H       ; Configure P3 as output (Control Signals)
5  ;=====
6  START:
7  ; Select Channel 0 (A=0, B=0, C=0)
8  CLR P2.0
9  CLR P2.1
10 CLR P2.2
11
12
13 ; Start ADC Conversion (Low-to-High Pulse on SC)
14 CLR P3.0           ; Ensure SC is LOW
15 NOP               ; Small delay
16 SETB P3.0         ; SC = 1 (Start Conversion)
17 LCALL SHORT_DELAY  ; Small delay to stabilize
18 CLR P3.0           ; SC = 0 (ADC starts conversion)
19
20 ;=====
21 ; Wait for End of Conversion (EOC)
22 WAIT_EOC:
23 JNB P3.1, WAIT_EOC ; Wait until EOC (P3.1) goes HIGH
24 LCALL SHORT_DELAY  ; Small delay for stability
25
26 ; Read ADC Data
27 SETB P3.2          ; OE = 1 (Enable ADC Output)
28 MOV A, P1          ; Read ADC value from Port 1
29 CLR P3.2           ; OE = 0 (Disable ADC Output)
30
31 ; Compare with 75% Threshold (191 in Decimal)
32 MOV B, #191
33 SUBB A, B
34 JNC BUZZER_ON      ; If ADC >= 191, turn on buzzer
35 CLR P3.5           ; Otherwise, keep buzzer OFF
36 SJMP START         ; Repeat process
37
38 BUZZER_ON:
39 SETB P3.5          ; Turn ON buzzer (P3.5)

```



```
litr40  ltrc  LCALL DELAY_150MS    ; Generate 150ms delay
litr41  ltrc  CLR P3.5         ; Turn OFF buzzer
litr42  ltrc  SJMP START      ; Repeat process
litr43  ltrc
litr44  ltrc  ;=====
litr45  ltrc  ; Subroutine for 150ms Delay (12 MHz Crystal)
litr46  ltrc  DELAY_150MS:
litr47  ltrc  MOV TMOD, #10H    ; Timer 1 Mode 1 (16-bit)
litr48  ltrc  MOV R2, #02       ; Repeat 2 times to achieve 150ms
litr49  ltrc
litr50  ltrc  REPEAT_DELAY:
litr51  ltrc  MOV TH1, #00H     ; Load Timer High Byte
litr52  ltrc  MOV TL1, #00H     ; Load Timer Low Byte
litr53  ltrc  SETB TR1         ; Start Timer 1
litr54  ltrc
litr55  ltrc  WAIT_TIMER:
litr56  ltrc  JNB TF1, WAIT_TIMER ; Wait for Timer Overflow
litr57  ltrc  CLR TR1          ; Stop Timer
litr58  ltrc  CLR TF1          ; Clear Overflow Flag
litr59  ltrc  DJNZ R2, REPEAT_DELAY ; Repeat for 150ms total
litr60  ltrc  RET
litr61  ltrc
litr62  ltrc  ;=====
litr63  ltrc  ; Short Delay Subroutine for ADC Stability
litr64  ltrc  SHORT_DELAY:
litr65  ltrc  MOV R5, #250
litr66  ltrc  SHORT_WAIT:
litr67  ltrc  DJNZ R5, SHORT_WAIT
litr68  ltrc  RET
litr69  ltrc
litr70  ltrc  END
```



SCHOOL OF ELECTRONICS ENGINEERING (SENSE)

Continuous Assessment Test – II

Winter Semester -2023-24

Programme Name & Branch : B.Tech ECE

Course Name & Code : BECE204L- Microprocessors and Microcontrollers

Slot: B1

Answer all the questions

Exam Duration: 50 Minutes

S.No	Question	Marks						
1	Assume a switch is connected to port P2.0, LED is connected to P1.1 and buzzer is connected to P2.2 of 8051 microcontroller. Write an assembly program for 8051 microcontroller to perform the following operations <table><tr><th>Input Switch (P2.0)</th><th>Output</th></tr><tr><td>0</td><td>Buzzer is ON for 15ms</td></tr><tr><td>1</td><td>LED is ON for 20ms</td></tr></table>	Input Switch (P2.0)	Output	0	Buzzer is ON for 15ms	1	LED is ON for 20ms	10
Input Switch (P2.0)	Output							
0	Buzzer is ON for 15ms							
1	LED is ON for 20ms							
2	Pin P1.0 of 8051 is connected to a switch. If the switch is closed (logic 0), then write an 8051 assembly language program to transmit "CLOSED" serially at the baud rate of 19200. If the switch is open (logic 1), then serially transmit "OPEN" at the baud rate of 9600. Assume "CLOSED" is stored in ROM location 200H onwards and "OPEN" in 300H location onwards.	10						
3	Write an assembly program using interrupt of 8051 microcontroller to simultaneously create 7 KHz and 500 Hz square waves on port pin P1.7 and P1.6.	10						
4.	8051 microcontroller is connected to water level sensor through ADC0808 which gives 5V when the water tank is 100% full. Develop an assembly program with suitable interfacing diagram to interface ADC0808 with 8051 to compare the water level stored in accumulator(A) with threshold value "E6H". If $A > E6H$ send a message 'HIGH LEVEL' through serial port at a baud rate of 2400. Assume 'HIGH LEVEL' is stored in ROM location 500H onwards.	10						



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5. Write an 8051 assembly program to interface 16x2 LCD display with suitable diagram. Assume the string "HELLO" stored in ROM location from 300H onwards. A switch is connected to port pin P2.7. If the switch input is 1, display the string in line 1 otherwise display the string in reverse order in line 2 as shown below.

If switch P2.7=1 display

						H	E	L	L	O					

If switch P2.7=0 display

						O	L	L	E	H					

10

**VIT**

Vellore Institute of Technology

School of Electronics Engineering**Winter Semester 2023-24 Continuous Assessment Test – II****Programme Name & Branch:** B. Tech (School of Computer Science Engineering)**SLOT:** C1+TC1**Course Name & Code:** Microprocessors and Microcontrollers BECE204L**Exam Duration:** 90 Min.**Maximum Marks:** 50

Q. No.	Questions	Max Marks	CO	BL
1	1. A) How the physical address is calculated and find it if segment address is 4000H and offset address is 3000H. Why are these two addresses required in 8086?	5	CO2	BL2
	B) Find the errors in the following instructions (if so). I. MOV DS,5000H II. ADD [AX], [4000H] III. ADD 0200H IV. INC 3000H V. CMP BX	5	CO2	BL2
2	Write an 8086 ALP programs to find the average of 64 numbers stored in the memory location 5000H with and without using DIV instruction. Assume that the sum does not exceed 16 bits.	10	CO2	BL4
3	Write an 8051 ALP to find (compute) the numbers which are divisible by both 3 and 7 in the range 1 to 99 and store them in memory. Interface two seven segment displays in P0 and P1 and display the stored numbers one after other.	10	CO2	BL4
4	Write an 8051 (AT89C51) ALP to generate a pulse waveform with frequency 2KHz using timer. Don't access TL0 or TL1 register in the ALP. Provide the necessary timer calculations.	10	CO4	BL4
5	Write 8051 (AT89C51) ALP for the following logic. The existing values of TCON=01H and IE = 0FFH. IF (P3.2 ==1) GENERATE WAVEFORM WITH 50% DUTY CYCLE. ELSE GENERATE WAVEFORM WITH 33% DUTY CYCLE.	10	CO4	BL4

Q1

How the physical address is calculated and find it if segment address is 4000H and offset address is 3000H. Why are these two addresses required in 8086?

$\text{PHY ADD} = \text{SEG ADD} * 10\text{H} + \text{OFFSET ADD}$

PHY ADD = 43000H

1MB of memory - interfaced with 8086 - requires 20-bit address lines.

8086 has 16-bit registers – 20-bit address is generated from two 16-bit address (seg address and offset address)

Find the errors in the following instructions (if so).

- | | |
|------------------------------|------------------------|
| I. MOV DS,5000H | ---- MOV DS, AX |
| II. ADD [AX], [4000H] | ---- ADD AX, [4000H] |
| III. ADD 0200H | ---- (or) ADD AX 0200H |
| IV. INC 3000H | ---- INC [3000H] |
| V. CMP BX | ---- CMP AX, BX |

Q2

Write an 8086 ALP programs to find the average of 64 numbers stored in the memory location 5000H with and without using DIV instruction. Assume that the sum does not exceed 16 bits.

i) WITH DIV

ASSUME CS:CODE, DS:DATA

DATA SEGMENT

NUMLIST DB 45H, 78H, 97H, 67H, 89H, 78H, 57H, 79H

NEXT DB 38H DUP (00H) // REMAINING VALUES WILL BE ZEROS

COUNT EQU 40H

RESULT DW 01H DUP (?)

DATA ENDS

CODE SEGMENT

START:

MOV AX, DATA ; get the segment address of DATA segment in DS register

MOV DS, AX

MOV CX, COUNT ; COUNT is the number of data has to be added

MOV SI, OFFSET NUMLIST ; get the offset address of NUMLIST

MOV AX, 0000H

```

MOV BX, 0000H
LP: MOV BL, [SI]
ADD AX, BX
INC SI
DEC CX
JNZ LP
MOV BX, 0040H
DIV BX
MOV DI, OFFSET RESULT ; get the offset address of RESULT in DI register
MOV [DI], AX
MOV AH, 00H
INT 21H
CODE ENDS
END START

RET

```

ii) WITHOUT DIV

```

ASSUME CS:CODE, DS:DATA

```

```

DATA SEGMENT

```

```

NUMLIST DB 45H, 78H, 97H, 67H, 89H, 78H, 57H, 79H

```

```

NEXT DB 38H DUP (00H) // REMAINING VALUES WILL BE ZEROS

```

```

COUNT EQU 40H

```

```

RESULT DW 01H DUP (?)

```

DATA ENDS

CODE SEGMENT

START:

MOV AX, DATA ; get the segment address of DATA segment in DS register

MOV DS, AX

MOV CX, COUNT ; COUNT is the number of data has to be added

MOV SI, OFFSET NUMLIST ; get the offset address of NUMLIST

MOV AX, 0000H

MOV BX, 0000H

LP: MOV BL, [SI]

ADD AX, BX

INC SI

DEC CX

JNZ LP

MOV CL, 06H

SHR AX, CL

MOV DI, OFFSET RESULT ; get the offset address of RESULT in DI register

MOV [DI], AX

MOV AH, 00H

INT 21H

CODE ENDS

END START

RET

Q3

Write an 8051 ALP to find (compute) the numbers which are divisible by both 3 and 7 in the range 1 to 99 and store them in memory. Interface two seven segment displays in P0 and P1 and display the stored numbers one after other.

ORG 0000H

; 7 SEGMENT LOOKUP TABLE

MOV 30H, #03FH

MOV 31H, #06H

MOV 32H, #5BH

MOV 33H, #04FH

MOV 34H, #066H

MOV 35H, #06DH

MOV 36H, #07DH

MOV 37H, #07FH

MOV 38H, #07FH

MOV 39H, #067H

MOV R0, #20H

MOV R2, #00H

MOV R3, #01H

LOOP:MOV A, R3

MOV B, #15H

DIV AB

MOV A, B

CJNE A, #00H,J1

MOV A, R3

MOV @R0, A

INC R0

INC R2

J1:INC R3

CJNE R3, #64H, LOOP

MOV R0, #20H

BACK: MOV A, @R0

MOV B, #100D

DIV AB

MOV A, B

MOV B, #10D

DIV AB

ADD A, #30H

MOV R1, A

MOV A, @R1

MOV P0, A

MOV A, B

```

ADD A, #30H
MOV R1, A
MOV A, @R1
MOV P1, A
INC R0
ACALL DELAY
DJNZ R2, BACK

```

```

ORG 100H
DELAY:
MOV R5, #255
H3: MOV R4, #255
H2: MOV R3, #255
H1: DJNZ R3, H1
    DJNZ R4, H2
    DJNZ R5, H3
RET
END

```

Q4

Write an 8051 (AT89C51) ALP to generate a pulse waveform with frequency 2KHz using timer. Don't access TL0/1 register in the ALP. Provide the necessary timer calculations.

Time period = $1/(2\text{KHz}) = 0.5\text{ms}$

ON period = OFF period = 0.25ms

Number of count = $0.25\text{ms}/1.085\mu\text{s} = 230.41 = 230$

TH0 = $255 - 230 = 25$ or 19H

```

ORG 0000H
LJMP MAIN ;by-pass interrupt vector table
ORG 000BH ;Timer 0 interrupt vector table
CPL P2.1 ;toggle P2.1 pin

```

RETI

ORG 0050H ;after vector table space
MAIN: MOV TMOD,#02H ;Timer 0, mode 2
MOV P0,#0FFH ;make P0 an input port
MOV TH0,#019H ;TH0=A4H for -92
MOV IE,#82H ;IE=10000010 (bin) enable
;Timer 0
SETB TR0 ;Start Timer 0
BACK: SJMP BACK ;get data from P0

END

Q5

Write 8051 (AT89C51) ALP for the following logic. The existing values of TCON=01H and IE = 0FFH.

ORG 0000H
LJMP MAIN ;by-pass interrupt
;vector table
;--ISR for INT1 to turn on LED
ORG 0003H ;INT0 ISR
SETB P1.3 ;turn on LED
MOV R3,#255
BACK: DJNZ R3,BACK ;keep LED on for a while
CLR P1.3 ;turn off the LED
MOV R3,#255
BACK1: DJNZ R3,BACK1
RETI ;return from ISR
;--MAIN program for initialization
ORG 30H
MAIN: MOV IE,#0FFH

```
MOV TCON, 01H;enable external INT 1
HERE: SETB P1.3 ;turn on LED
MOV R3,#255
BACK4: DJNZ R3,BACK4 ;keep LED on for a while
CLR P1.3 ;turn off the LED
MOV R3,#255
BACK3: DJNZ R3,BACK3
MOV R3,#255
BACK2: DJNZ R3,BACK2;stay here until get interrupted
SJMP HERE
END
```



School of Electronics Engineering

Winter Semester 2023-24 Continuous Assessment Test – II

Programme Name & Branch: B. Tech (School of Computer Science Engineering)

SLOT: C2+TC2

Course Name & Code: Microprocessors and Microcontrollers- BECE204L

Exam Duration: 90 Min.

Maximum Marks: 50

Q. No.	Questions	Max Marks
1	a) CS = 4000H and DS = 4500H. Find if there is any overlap between these two segments. If so, what is the size of the overlap.	5
	b) Find the errors in the following instructions (if so). i. MOV CS,5000H ii. SBB [AX], [2000H] iii. MUL AL, BL iv. AL =83H; CL=29H; ADD AL, CL; DAA; After execution (AL)=05, carry flag (CY)=0; v. OR AX, 0098H contents of AX = 3F0FH, after execution AX = 3090H.	5
2	Write an 8086 ALP to find (compute) the values of the function $f(x) = x^2 + 3x + 2$ in the range $x=1$ to $x=10$ and store them in the memory location from 5000H onwards.	10
3	Write an 8051 ALP to multiply two numbers without using MUL AB instruction and assume that the product does not exceeds 8-bits. Display the digits of the product on the seven segment displays connected to P0 and P1.	10
4	Write an 8051 (AT89C51) ALP to generate a periodic waveform simultaneously as follows: i) 0.1ms ON & 0.1ms OFF ii) 0.2ms ON & 0.2ms OFF using timer. Don't access TL0 or TL1 register in the ALP. Provide necessary timer calculations.	10
5	Write 8051 (AT89C51) ALP for the following logic. The existing values of TCON=01H and IE = 0FFH. IF (P3.3 =1) SEND THE SEQUENCE 0,1,2,3,5,6,7,8,9 REPEATLY IN P0. ELSE SEND THE SEQUENCE 9,8,7,6,5,4,3,2,1 REPEATLY IN P1.	10



School of Computer Science Engineering

Winter Semester 2023-2024

Continuous Assessment Test – II

Program Name & Branch: BTECH CSE (BCB, BCE, BCI, BCT, BDS, BKT)

Course Name & code: BECE204L Microprocessors and Microcontrollers

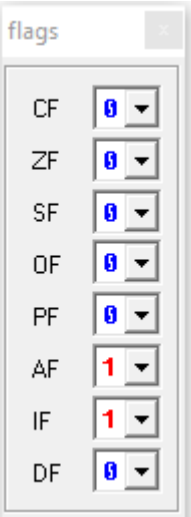
Slot: F1+TF1

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s):

Answer all the questions

S. No	Question	Marks
1. A	Write the physical address of memory which is interfaced with 8086. The logical address is 1000:0200 H. Ans: Physical Address = (10H X 1000H) + 0200H = 10200H	2
B	Write the status of flag register of 8086 after executing the following program: MOV AX, 00FFH MOV BX, 00F2H ADD AX, BX Ans: CF = 0, ZF = 0, SF = 0, OF = 0, PF = 0, AF = 1, IF = 1, DF = 0.	3
		5
C	Identify the status of the Accumulator after executing each line of the following program: MOV AL,23H // AL = 23H MOV BL,45H //AL = 23H 0010 0011 ADD AL,BL //AL = 68H 0110 1000 XOR AL,BL //AL =2DH 0110 1000 XOR 0100 0101 = 0010 1101 = 2DH SUB AL,3H // 2AH MOV BL,AL //2AH	

2	Develop an 8086 ALP for $(A + B)^2$. Assume A and B values are 0 – 9 only. Ans: Assume (i) Data A is stored logical address at 3000H, Data B is stored at 3001H, (ii) A and B values are 5 and 6 respectively. <pre> MOV [3000H], 05H MOV [3001H], 06H MOV AL,[3000H] ADD AL,[3001H] MUL AL HLT </pre>	10
3	With the help of interrupt programming, develop an ALP of 8051 microcontrollers to perform the following tasks; continuously get numeric data from P0 and send it to P1. If the data is even, serially send it through port P2 with a baud rate of 19200. Otherwise, the data will be sent through P2 with a baud rate of 9600. Ans: <pre> ORG 0000H LJMP MAIN ORG 0023H LJMP SERIAL ORG 0040H MAIN: MOV TMOD,#20H // Mode 2 of Timer 1 (Auto reload mode) MOV TH1,#-3 // Baud rate MOV SCON,#50H // Mode 1 in Serial communication MOV IE, #10010000B // Enable Serial Communication Interrupt SETB TR1 // start the timer XX: JB P2.0, L5 MOV PCON,#00H //For odd number , Baud rate = 9600, PCON = 00H SJMP L6 L5: MOV PCON,#80H //For even number , Baud rate = 19200, PCON = 80H SJMP L6 L6: MOV P1,P0 //copy data from Port 0 to Port1 continuously SJMP XX SERIAL: CLR TI MOV SBUF,P2 //copy data from Port 2 to UART continuously RETI END </pre>	10
4	A football stadium is designed with a seating capacity of precisely 1000 persons. It is assumed that there is just one entry and exit gate accessible within the stadium. The entry gate is configured so that whenever a person enters a stadium, a high-to-low pulse will be generated and that will be given to the port3.4 of the 8051 microcontroller. The stepper motor regulates the opening and shutting of the entry gate. To open/close the entry gate, send a logic 1/logic 0 to	10

	port0.5 of the 8051 microcontroller, which connects to the stepper motor. Develop an ALP of 8051 that counts and allows maximum up to 1000 persons and closes the gate after that irrespective of persons leaving from the exit gate. Use the mode 2 timer programming. Ans: <pre> ORG 0000H MOV TMOD, #06H // TIMER-0 AS A COUNTER MOV TH0, #00 SETB P0.5 // Opening of the GATE L2: SETB TR0 L1: JB TF0, L4 MOV A, TL0 MOV P1, A CJNE R1, #03h, L1 // LOGIC FOR CHECKING THE 1000 PERSONS L3: CJNE A, #0E8h, L1 // LOGIC FOR CHECKING THE 1000 PERSONS CLR P0.5 // CLOSING OF THE GATE SJMP Here L4: CLR TR0 CLR TF0 INC R1 SJMP L2 Here: SJMP Here END </pre>																															
5	Develop an ALP of 8051 for displaying reversely the nine characters string (Your Reg. No) on one line LCD display with Cursor blinking. The HEX value of command words are given below. <table border="1"> <thead> <tr> <th>S. No</th><th>HEX Value</th><th>Command to LCD</th></tr> </thead> <tbody> <tr> <td>1.</td><td>38H</td><td>8-bit 2-line 5x7 Dots LCD</td></tr> <tr> <td>2.</td><td>30H</td><td>8-bit 1-line 5x7 Dots LCD</td></tr> <tr> <td>3.</td><td>06H</td><td>Increment cursor</td></tr> <tr> <td>4.</td><td>04H</td><td>Decrement cursor</td></tr> <tr> <td>5.</td><td>80H</td><td>Beginning of First Row</td></tr> <tr> <td>6.</td><td>8FH</td><td>End of First Row</td></tr> <tr> <td>7.</td><td>0EH</td><td>Display on Cursor on</td></tr> <tr> <td>8.</td><td>0FH</td><td>Display on Cursor blinking</td></tr> <tr> <td>9.</td><td>01H</td><td>Clear cursor</td></tr> </tbody> </table> Assume Port2 is used for Data/Command word transferring, Port1.0 is connected to RS of LCD, Port1.1 is connected to $\overline{R/\overline{W}}$ of LCD and Port1.2 is connected to Enable of LCD. Ans: Suppose Reg. No is 22BCE0123 <pre> ORG 0H MOV A, #30H ACALL CMDWRT ACALL DELAY MOV A, #0FH </pre>	S. No	HEX Value	Command to LCD	1.	38H	8-bit 2-line 5x7 Dots LCD	2.	30H	8-bit 1-line 5x7 Dots LCD	3.	06H	Increment cursor	4.	04H	Decrement cursor	5.	80H	Beginning of First Row	6.	8FH	End of First Row	7.	0EH	Display on Cursor on	8.	0FH	Display on Cursor blinking	9.	01H	Clear cursor	10
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9.	01H	Clear cursor																														

ACALL CMDWRT
ACALL DELAY

MOV A,#01H
ACALL CMDWRT
ACALL DELAY

MOV A,#04H
ACALL CMDWRT
ACALL DELAY

MOV A,#8FH
ACALL CMDWRT
ACALL DELAY

MOV A,#'2'
ACALL DATAWRT
ACALL DELAY

MOV A,#'2'
ACALL DATAWRT
ACALL DELAY

MOV A,#'B'
ACALL DATAWRT
ACALL DELAY

MOV A,#'C '
ACALL DATAWRT
ACALL DELAY

MOV A,#'E'
ACALL DATAWRT
ACALL DELAY

MOV A,#'0'
ACALL DATAWRT
ACALL DELAY

MOV A,#'1'
ACALL DATAWRT
ACALL DELAY

MOV A,#'2'
ACALL DATAWRT
ACALL DELAY

MOV A,#'3'
ACALL DATAWRT
ACALL DELAY

AGAIN: SJMP AGAIN

CMDWRT:MOV P2,A
CLR P1.0

	CLR P1.1 SETB P1.2 ACALL DELAY CLR P1.2 RET DATAWRT:MOV P2,A SETB P1.0 CLR P1.1 SETB P1.2 ACALL DELAY CLR P1.2 RET DELAY:MOV R3,#40 HERE2: MOV R4,#255 HERE:DJNZ R4,HERE DJNZ R3,HERE2 RET END	
--	---	--

(b) For the given 8051 assembly program, find out the time required to execute it, if XTAL frequency is 11.0592 MHz.

MOV R4, #0FH	1 machine cycle
L1: ADD A, R4	1 machine cycle
MOV B, #89H	1 machine cycle
MUL A B	2 machine cycle
INC A	1 machine cycle
DJNZ R4, L1	2 machine cycle

Final Assessment Test – April 2025

Course: BECE204L - Microprocessors and Microcontrollers

Class/ NBR(s): 4038 / 4040 / 4042 / 4045 / 4047 / 4049 /
4051 / 4054 / 4057 / 4059 / 4061 / 4063 /
4065 / 4068 / 4071 / 4073 / 4075 / 4077 /
4078 / 4080 / 4082 / 4084 / 4086 / 4088 /
4191 / 4199

Slot: B1+TB1

Time: Three Hours

Max. Marks: 100

- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
➤ DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions

(10 X 10 = 100 Marks)

1. Explain the concept of interrupt handling in an 8-bit embedded system. How does the processor manage multiple interrupts, and what is the significance of interrupt prioritization in embedded systems?
2. (i) The 8086 microprocessor has a 16-bit data bus and a 20-bit address bus. How does this affect the addressing capability, and what limitations does this impose on memory access?
(ii) In 8086 Microprocessor, demonstrate the content of FLAGS after performing $(FFFF)_{16} + (0001)_{16}$
3. Develop an 8051 Assembly Language Program (ALP) to count the number of high bits (1s) of 8 bit data stored at RAM location 30H. Store the count in 31H.
4. For the following 8051 ALP, list the sequences of actions taking place while executing the entire program. Demonstrate your answer using proper sketch of Program Counter.

ROM LOCATION	INSTRUCTION
0000H	ORG 0000H
0001H	START:
0002H	MOV SP, #70H
0004H	MOV R0, #05
0005H	MOV R1, #00
0006H	LCALL CALC_SUM
0009H	HERE: SJMP HERE
0100H	CALC_SUM:
0101H	MOV A, R1
0102H	ADD A, R0
0103H	MOV R1, A
0104H	DJNZ R0, CALC_SUM
0106H	MOV B, R1
0107H	RET

5. Develop an 8051 ALP in which 8051 gets data from P1 and sends it to P2 continuously while incoming data from the serial port is sent to P0. Assume that XTAL=11.0592 MHz. Set the baud rate at 9600.

6. Assume there are two switches connected to MSB and LSB pins of Port 0 of 8051 Microcontroller. Develop an ALP that continuously monitors the MSB and LSB pins of P0 and based on the binary combinations; perform the actions as given in table 1.

Table1

S.No	MSB	LSB	Action
1	0	1	Generate a waveform of frequency 5KHz at P1.0 use Timer 0
2	1	0	Transmit character "A" continuously at the baud rate of 19200

7.(a) Develop an 8051 assembly code to display the string "HELLO" on a common cathode 7-segment display with a 500ms interval between each character. Assume that the pattern for "H", "E", "L", "L", "O" are stored in ROM starting at 400H. Use Timer 1 to generate 500ms of delay.

OR

7.(b) Develop an 8051 ALP to display a sequence of numbers on a 16x2 LCD. Assume the numbers (e.g., "1", "2", "3", ...till "9") are stored in ROM starting at 800H as ASCII values. The program should display these numbers sequentially on the first row of the LCD, with each number appearing for 1 second before the next number is displayed. Use Timer 0 to generate the delay.

Reference:

01 – clear display

38 – 2 lines 5X7 matrix

.0E – Display ON cursor blinking

06 – Increment cursor

04 – Decrement cursor

80 – Force cursor to beginning of first line.

C0 - force cursor to beginning of Second line

- 8.(a) Draw the diagram for interfacing 8051 with DAC 0808 at Port P1. Develop 8051 assembly programs to generate (i) a stair-step ramp waveform and (ii) a triangular waveform.

OR

- 8.(b) Assume a pH sensor is used to measure the soil pH level and is interfaced with an 8051 microcontroller. Design the block diagram and a circuit diagram for the interfacing. The sensor provides an analog output corresponding to the pH value. Assume step size of 10mV. Explain the necessary signal conditioning and how the 8051 processes the sensor's data to display the pH level.
9. (i) The ARM core follows the RISC (Reduced Instruction Set Computing) architecture, which emphasizes delivering simple yet efficient instructions. How do the four key design principles of RISC architecture contribute to its performance in real-time applications, such as smartphones or embedded systems?
- (ii) Describe the different processor modes in ARM processors and explain the corresponding register banks associated with each mode. Use diagrams to illustrate how these modes operate, highlighting how each mode facilitates efficient processing and system management.
10. (i) Explain with examples the different addressing modes for single register load-store ARM instructions.
- (ii) Explain the effect of barrel shifter on the following instructions in an ARM processor. What is the result of the Logical Shift Left operation on these instructions?
- (a) MOV r7, r5, LSL #2
- (b) ADD r0, r1, r1 LSL #1

⇔⇔⇔ W/D/TY ⇔⇔⇔


VIT

 Vellore Institute of Technology
(Approved as a University under section 3 of the UGC Act, 1956)
Final Assessment Test - April 2025

 Course: **BECE204L - Microprocessors and Microcontrollers**

 Class NBR(s): **4037 / 4039 / 4041 / 4043 / 4046 / 4048 /**
4050 / 4053 / 4055 / 4058 / 4060 / 4062 /
4064 / 4067 / 4069 / 4072 / 4074 / 4076 /
4081 / 4083 / 4085 / 4087 / 4089 / 4194

 Slot: **B2+TB2**

 Time: **Three Hours**

 Max. Marks: **100**

- **KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE**
- **DON'T WRITE ANYTHING ON THE QUESTION PAPER**

 Answer ALL Questions

(10 X 10 = 100 Marks)

1. In an 8-bit embedded system, how does the stack pointer and program counter interact during a subroutine call and return? What problems might arise if the stack overflows or underflows?
2. (i) Discuss the concept of "pipelining" in the 8086 microprocessor. How does the 8086 handle instruction prefetching, and what impact does this have on its performance?
(ii) In 8086 Microprocessor, demonstrate the content of AX register and FLAGS after performing ADD AX, BX
Given: AX = 0x8000 and BX = 0x8000
3. Develop an 8051 Assembly Language Program (ALP) to compare three 8-bit numbers stored in memory locations 30H, 31H, and 32H. Store the largest number at location 33H.
4. Analyse the following program and specify the data in the RAM locations, register A and Stack Pointer [SP] after execution of the program.

```

ORG 0
MOV SP, #60H
MOV R0, #10H
MOV R1, #20H
MOV R2, #30H
MOV R3, #40H
MOV R4, #50H
MOV R5, #60H
MOV R6, #70H
MOV R7, #80H
PUSH 02
  
```

```

PUSH 03
MOV A, R1
SUBB A, R0
JZ SkipPop
POP 05
SkipPop:
PUSH 04
MOV A, R6
ADD A, R5
POP 06
ADD A, R7
END

```

- END
5. ✓ Develop an 8051 ALP to generate a wave of frequency 2 KHz with 50% duty cycle on P1.0 using Timer 1 for the delay creation. Assume XTAL = 11.0592MHz while P1 reads data from external source which should be sent to P2 continuously.
(change use timer)
 6. ✓ Develop an 8051 microcontroller ALP in which 8051 gets incoming data from the serial port. If the received data is odd then turn ON an LED at P1.3 else sound a buzzer at P2.4.
 7. (a) ✓ Develop an 8051 ALP to store the message "WELCOME TO 8051" starting at ROM location 0300H. Display the characters on the LCD as show in the figure below

[illegible]

Reference:

01 – clear display

38 – 2 lines 5X7 matrix

OE – Display ON cursor blinking

06 – Increment cursor

04 – Decrement cursor

80 - Force cursor to beginning of first line.

C0 - force cursor to beginning of Second line

OR

- 7.(b) Develop an 8051 assembly code to display a countdown from "3" to "0" on a common cathode 7-segment display. The program should create a 1-second interval between each digit. Use Timer 1 to generate 1-second delay.
- 8.(a) Assume a switch is connected to P0.7. If SW=0, develop a program to generate a ramp waveform of 10 steps by interfacing DAC0808 with 8051 microcontroller else generate a saw tooth waveform with 5 steps.

OR

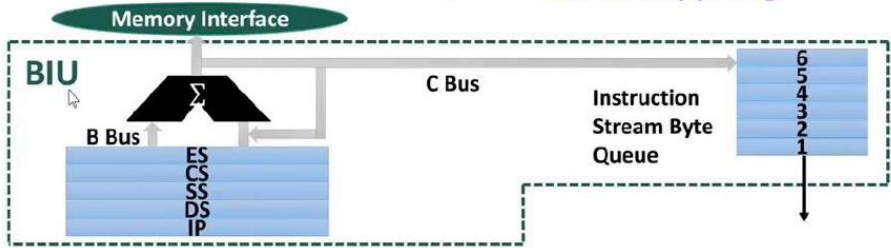
- 8.(b) Design a block diagram and the circuit diagram to interface a Heart Rate Monitor sensor with the 8051 microcontroller. The sensor provides an analog signal corresponding to the heartbeat rate. Explain the necessary signal conditioning steps, and describe how the microcontroller processes the data to display the heart rate on an LCD.
9. How does the data flow between the ALU (Arithmetic Logic Unit) and the Barrel Shifter in the ARM processor improve performance in real-time applications, such as mobile devices or IoT systems? Explain the process with a diagram, highlighting how efficient data manipulation and shifting operations are used in real-time control systems.
10. Apply the ARM instruction set to
- (i) Move an 8-bit immediate value into R7.
 - (ii) Add a 16-bit immediate value to the contents of R5 and store the result in R5.
 - (iii) Logical shift left the contents of R8 by 4 and store the result in R6.
 - (iv) Subtract a 16-bit immediate value from the contents of R2 and store the result in R2.
 - (v) Compare the contents of R10 with a 16-bit immediate value.

⇒⇒⇒ X/D/TY ⇒⇒⇒

Final Assessment Test- KEY – Winter 2023-24 Semester - May 2024

Course code : BECE204L	Slot: C1+TC1
Course Title : Microprocessors and Microcontrollers	Time: Three Hours
Emp Id: shanidul.hoque@vit.ac.in	Course Mode : CBL
+91-9401408475	Max. Marks: 100
Faculty Name : Dr. Shanidul Hoque	School: SENSE

Sl. No:	Class Number(s): VL2023240505477																																		
1.	What are the differences between i3, i5, and i7 processors, and how do they compare in terms of performance, features, and suitability for various computing tasks? ANS. <table border="1"> <thead> <tr> <th>Intel Core i3</th><th>Intel Core i5</th><th>Intel Core i7</th></tr> </thead> <tbody> <tr> <td>Intel Core i3 is an entry-level processor design for low-end applications.</td><td>Intel Core i5 is a mid-range processor developed for everyday use computer systems.</td><td>Intel Core i7 is a high-end processor designed for applications that need high processing power.</td></tr> <tr> <td>Core i3 processor is a dual-core processor, i.e. it has two physical cores.</td><td>Core i5 is available in quad-core (2 cores) and quad core (4 cores).</td><td>Core i7 is available in quad core (4 cores), six cores, and eight-cores.</td></tr> <tr> <td>Core i3 has a cache of size either 3 MB or 4 MB.</td><td>Core i5 has a cache of size ranging between 3 MB and 8 MB.</td><td>Core i7 has a cache of size ranging between 4 MB and 8 MB.</td></tr> <tr> <td>The clock speed of Core i3 processor is ranging between 1.30 GHz and 3.50 GHz.</td><td>The clock speed of Core i5 processor is ranging between 1.90 GHz and 3.80 GHz.</td><td>The clock speed of Core i7 processor is ranging between 2.6 GHz and 3.7 GHz.</td></tr> <tr> <td>Core i3 processor supports 4 threads.</td><td>Core i5 processor also supports 4 threads.</td><td>Core i7 processor supports 8 threads.</td></tr> <tr> <td>Core i3 processor does not support turbo boost feature.</td><td>Core i5 supports turbo boost.</td><td>Core i7 processor also supports turbo boost.</td></tr> <tr> <td>Core i3 processor does not support virtualization technology.</td><td>Core i5 processor supports virtualization technology.</td><td>Core i7 processor also supports virtualization technology.</td></tr> <tr> <td>Core i3 processor does not support Intel vPro technology.</td><td>Core i5 processor does not support Intel vPro technology.</td><td>Core i7 processor supports Intel vPro technology.</td></tr> <tr> <td>Core i3 processor is cheapest among the three.</td><td>Core i5 is mid-range processor.</td><td>Core i7 is expensive.</td></tr> <tr> <td>Core i3 processor is suitable to use for entry level tasks such as word processing, browsing, documentation, watching videos, etc.</td><td>Core i5 processor is suitable for basic work like web browsing, word processing, spreadsheets, basic graphics designing, etc.</td><td>Core i7 processor is suitable for high end applications, such as gaming, graphics designing, video editing, simulation, etc.</td></tr> </tbody> </table>	Intel Core i3	Intel Core i5	Intel Core i7	Intel Core i3 is an entry-level processor design for low-end applications.	Intel Core i5 is a mid-range processor developed for everyday use computer systems.	Intel Core i7 is a high-end processor designed for applications that need high processing power.	Core i3 processor is a dual-core processor, i.e. it has two physical cores.	Core i5 is available in quad-core (2 cores) and quad core (4 cores).	Core i7 is available in quad core (4 cores), six cores, and eight-cores.	Core i3 has a cache of size either 3 MB or 4 MB.	Core i5 has a cache of size ranging between 3 MB and 8 MB.	Core i7 has a cache of size ranging between 4 MB and 8 MB.	The clock speed of Core i3 processor is ranging between 1.30 GHz and 3.50 GHz.	The clock speed of Core i5 processor is ranging between 1.90 GHz and 3.80 GHz.	The clock speed of Core i7 processor is ranging between 2.6 GHz and 3.7 GHz.	Core i3 processor supports 4 threads.	Core i5 processor also supports 4 threads.	Core i7 processor supports 8 threads.	Core i3 processor does not support turbo boost feature.	Core i5 supports turbo boost.	Core i7 processor also supports turbo boost.	Core i3 processor does not support virtualization technology.	Core i5 processor supports virtualization technology.	Core i7 processor also supports virtualization technology.	Core i3 processor does not support Intel vPro technology.	Core i5 processor does not support Intel vPro technology.	Core i7 processor supports Intel vPro technology.	Core i3 processor is cheapest among the three.	Core i5 is mid-range processor.	Core i7 is expensive.	Core i3 processor is suitable to use for entry level tasks such as word processing, browsing, documentation, watching videos, etc.	Core i5 processor is suitable for basic work like web browsing, word processing, spreadsheets, basic graphics designing, etc.	Core i7 processor is suitable for high end applications, such as gaming, graphics designing, video editing, simulation, etc.	10
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Core i3 processor is cheapest among the three.	Core i5 is mid-range processor.	Core i7 is expensive.																																	
Core i3 processor is suitable to use for entry level tasks such as word processing, browsing, documentation, watching videos, etc.	Core i5 processor is suitable for basic work like web browsing, word processing, spreadsheets, basic graphics designing, etc.	Core i7 processor is suitable for high end applications, such as gaming, graphics designing, video editing, simulation, etc.																																	

2.	(a) What are the key architectural components of the Intel 8086 microprocessor? Explain the function of Bus Interface Unit (BIU) with suitable diagram. ANS. The key architectural components of the Intel 8086 microprocessor are Bus Interface Unit (BIU): Instruction queue, Segmentation register, Instruction pointer Execution Unit (EU): Control Unit, Registers, ALU, Flag Register Explain each block of BIU:  (b) Rectify errors if any and identify the addressing mode of each of the following instruction in 8086 microprocessor: (i) MOV AX, #3FE1H (ii) MOV DS, 3000H (iii) MOV BL, CX (iv) MOV AX, 50H[BX] (v) MOV CL, [BX+SI+30H] ANS: (i) Error; MOV AX, 3FE1H ; Immediate addressing mode (ii) Error; MOV AX, 3000H; Immediate addressing MOV DS, AX; Register addressing (iii) Error; MOV BL, [CX]; Register indirect addressing (iv) No error; Register relative addressing (v) No error; Base relative indexed addressing	5+5 =10
3.	Develop an Intel 8086 assembly language program to count the odd and even numbers from a given list of 100 numbers stored at starting address of 2000:0300H and save the count values at 3000:0500H and 4000:0700H, respectively. ANS: //Comments should be written corresponding to the instructions <pre> MOV AX,2000H MOV DS,AX MOV SI,300H MOV CX,64H MOV BL,00H MOV BH,00H BACK: MOV AL,[SI] ROR AL,1 JC ODD INC BL JMP NEXT ODD: INC BH </pre>	10

```

NEXT: INC SI
      LOOP BACK
      MOV AX,3000H
      MOV DS,AX
      MOV [500H], BH
      MOV AX,4000H
      MOV DS,AX
      MOV [700H], BL
      HLT

```

4. (a) Assume a data (x) is present in Accumulator. If x is negative number, write an 8051 assembly language program to transfer a string of data from code space starting at address 200H to RAM location starting at 40H in reverse order. The data is given below:

10

0200H: DB "Believe in yourself "

ANS:

//Comments should be written corresponding to the instructions

Assume X is present in Accumulator

ORG 0000H

MOV A, #0F3H; Assume any 8-bit value of X , here e.g X=0F3H is considered.

RLC A

JNC BYPASS

MOV A, #00H

MOV DPTR, #0213H

MOV R1, #19

MOV R0, #40H

LOOP: CLR A

MOVC A,@A+DPTR

MOV @R0, A

DEC DPL

INC R0

DJNZ R1, LOOP

HERE: SJMP HERE

ORG 0200H

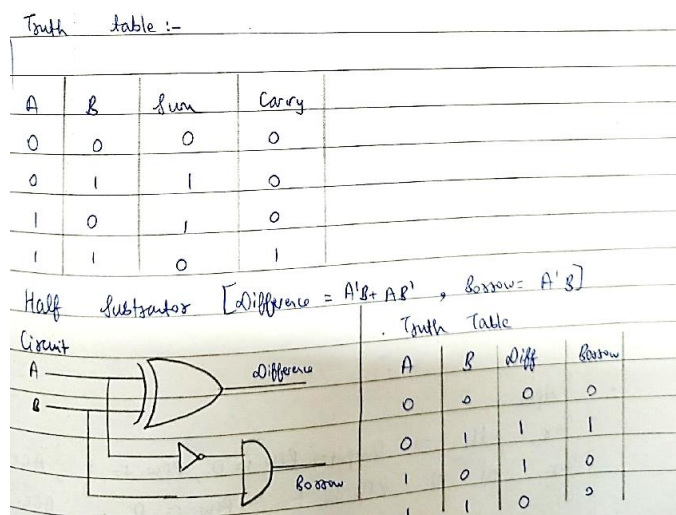
DB "VIT UNIVERSITY"

BYPASS: END

- (b) Develop an 8051 based assembly language program to implement the expressions for half adder and half subtractor circuits. Store the sum/Difference, carry and borrow in PSW.1, PSW.5 and ACC.7, respectively. (Use Basic Logic Gates only)

10

ANS.



```

ORG 0000H
SETB ACC.0 ; Input A
SETB ACC.1 ; Input B
MOV C, ACC.0 ; A
CPL C ;  $\bar{A}$ 
ANL C, ACC.1 ;  $\bar{A}B$ 
MOV ACC.7, C ;  $ACC.7 = \bar{A}B$ 
MOV C, ACC.1 ; B
CPL C ;  $\bar{B}$ 
ANL C, ACC.0 ;  $A\bar{B}$ 
ORL C, ACC.7 ;  $\bar{A}B + A\bar{B} = Sum$ 
MOV C, ACC.0 ; A
ANL C, ACC.1 ;  $AB = Carry$ 
END
MOV C, ACC.0 ; A
CPL C ;  $A'$ 
ANL C, ACC.1 ;  $A'B$ 
MOV ACC.0, C ;  $ACC.0 = A'B$  (Borrow)
END

```

5. Calculate the amount of delay caused by the delay subroutine if the system has an 8051 with frequency of 16 MHz. The number of machine cycles are provided in parentheses

```

DELAY: MOV R2, # 10010110B      (1)
AGAIN: MOV R3, #250              (1)
HERE:  NOP                      (1)
        NOP                      (1)
        NOP                      (1)
        DJNZ R3, HERE            (2)
        DJNZ R2, AGAIN          (2)
        RET                     (2)

```

ANS:

Time period of machine cycle = $12/16\text{MHz} = 0.75 \text{ microsec}$

Delay for HERE loop, $5 \times 250 \times 0.75 = 937.5 \text{ microsec}$

Delay for Again loop, $3 \times 150 \times 0.75 + 150 \times 937.5 = 140962.5 \text{ microsec}$

Total Delay = $3 \times 0.75 + 140962.5 = 140964.75 \text{ microsec}$

10

6.	Assume P1 and P2 of the 8051 are connected to LEDs and switches, respectively as shown in Figure 1. Write an 8051 assembly language program to (a) get data on switches connected to P2 and send it to the PC COM serially, (b) receive any data sent by the PC COM and put it on LEDs connected to P1. Perform both the tasks one after another continuously with 2400 baud rate. Figure 1 ANS: //Calculation of TH1 values //Comments should be written corresponding to the instructions <pre> ORG 0000H MOV P2,#0FFH ;make P2 an input port MOV TMOD,#20H ;Timer 1, mode 2(auto-reload) MOV TH1,#-12 ;2400 baud rate (or F4H) MOV SCON,#50H ;8-bit,1 stop, REN enabled SETB TR1 ;start Timer 1 AGN: MOV A,P2 ;read data on P2 ACALL SEND ;transfer it serially ACALL RECV ;get the serial data MOV P1,A ;display it on LEDs SJMP AGN ;stay in loop indefinitely ;-----serial data transfer. ACC has the data SEND: MOV SBUF,A ;load the data HERE: JNB TI, HERE ;stay here until last bit gone CLR TI ;get ready for next char RET ;return to caller ;-----receive data serially in ACC RECV: JNB RI, RECV ;wait here for char MOV A, SBUF ;save it in ACC CLR RI ;get ready for next char RET ;return to caller </pre>	10
7.	Write an 8051 assembly language program that continuously get 8-bit data from P0 and sends it to P1 while the P3.3 (INT1) pin is connected to a switch which is normally high. Whenever it goes low, it should turn on an LED for 100 microsecond which is connected to P2.5 pin. ANS: //Calculation of TH0 values //Comments should be written corresponding to the instructions <pre> ORG 0000H LJMP MAIN ;.....ISR..... ORG 0013H LJMP NEXT </pre>	10

	<pre> ;.....MAIN Program..... ORG 0030H MAIN: MOV P0,#0FFH MOV P1,#00H CLR P2.5 MOV IE, #10000100B HERE: MOV A,P0 MOV P1,A SJMP HERE ; NEXT:SETB P2.5 MOV TMOD, #02H MOV TH0, #-92 SETB TR0 L1: JNB TF0, L1; CLR TR0 CLR TF0 CLR P2.5 RETI END </pre>	
8.	(a) Write an 8051 assembly language program to interface an 16x2 LCD by sending Hex code commands “38H, 0EH, 01H, 06H” and display “VIT Vellore” on it in line 2 position 3. Assume D0-D7, RS, R/W’, and E pins of LCD are connected with P1.0-P1.7, P2.0, P2.1, and P2.2 pins of 8051 microcontroller, respectively. ANS: <pre> ORG 0000H MOV A,#38H ACALL COMNWRT ACALL DELAY MOV A,#0EH ACALL COMNWRT ACALL DELAY MOV A,#01H ACALL COMNWRT ACALL DELAY MOV A,#06H ACALL COMNWRT ACALL DELAY MOV A,#C3H ACALL COMNWRT ACALL DELAY </pre> <pre> D1: MOV DPTR,#MYDATA CLR A MOVC A,@A+DPTR ACALL DATAWRT ;call command subroutine ACALL DELAY ;give LCD some time INC DPTR JZ AGAIN SJMP D1 AGAIN: SJMP AGAIN ;stay here COMNWRT: ;send command to LCD MOV P1,A ;copy reg A to P1 CLR P2.0 ;RS=0 for command CLR P2.1 ;R/W=0 for write SETB P2.2 ;E=1 for high pulse ACALL DELAY ;give LCD some time CLR P2.2 ;E=0 for H-to-L pulse RET DATAWRT: ;write data to LCD MOV P1,A ;copy reg A to port 1 SETB P2.0 ;RS=1 for data CLR P2.1 ;R/W=0 for write SETB P2.2 ;E=1 for high pulse ACALL DELAY ;give LCD some time CLR P2.2 ;E=0 for H-to-L pulse RET DELAY: MOV R3,#250 ;50 or higher for fast CPUs HERE2: MOV R4,#255 ;R4 = 255 HERE: DJNZ R4,HERE ;stay until R4 becomes 0 DJNZ R3,HERE2 RET ORG 300H MYDATA: DB "VIT Vellore", 0 END </pre>	10

(b) Develop an assembly language program to interface a 4x4 matrix keyboard with 8051 microcontroller.

10

ANS.

```
MOV P2, #0FFH ; make P2 input (column)
L1: MOV P1, #00H ; make P2 output (row)
    MOV A, P2 ; read all column
    ANL A, #0FH;
    CJNE A, #0FH, L2
    SJMP L1

L2: ACALL Delay ; call 20 ms delay

    MOV A, P2 ;
    ANL A, #0FH ; check if any key is pressed
    CJNE A, #0FH, ScanRow ; key pressed
    SJMP L1
```

scan row

```
ScanRow: MOV P1, #1111110B ; ground row 0
    MOV A, P2; read all columns
    ANL A, #0FH
    CJNE A, #0FH, Row_0; key row 0, find column
    MOV P1, #11111101B ; ground row 1
    MOV A, P2
    ANL A, #0FH
    CJNE A, #0FH, Row_1; key row 1, find column
    MOV P1, #111111011B ; ground row 2
    MOV A, P2
    ANL A, #0FH
    CJNE A, #0FH, Row_2; key row 2, find column
    MOV P1, #1111110111B ; ground row 3
    MOV A, P2
    ANL A, #0FH
    CJNE A, #0FH, Row_3 ; key row 3, find column
    LJMP L1
```

```
Row_0: MOV DPTR, #KCODE0 ; set DPTR = start of Row 0
    SJMP FIND ; Find column key belongs to
Row_1: MOV DPTR, #KCODE1
    SJMP FIND
Row_2: MOV DPTR, #KCODE2
    SJMP FIND
Row_3: MOV DPTR, #KCODE3
    SJMP FIND

FIND: RRC A ; Check any carry bit is low
    JNC MATCH ; if zero get ASCII code
    INC DPTR
    SJMP FIND ; keep searching

MATCH: CLR A
    MOVC A, @A+DPTR ; get ASCII from table
    MOV P0, A ; display pressed key
    LJMP L1
```

ORG 300H

```
KCODE0: DB '0','1','2','3' ;ROW 0
KCODE1: DB '4','5','6','7' ;ROW 1
KCODE2: DB '8','9','A','B' ;ROW 2
KCODE3: DB 'C','D','E','F' ;ROW 3
```

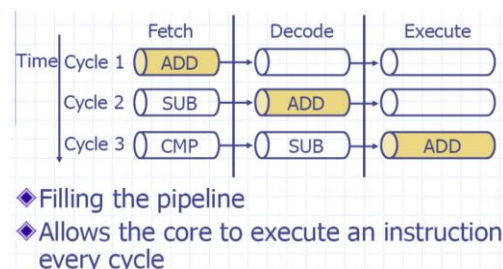
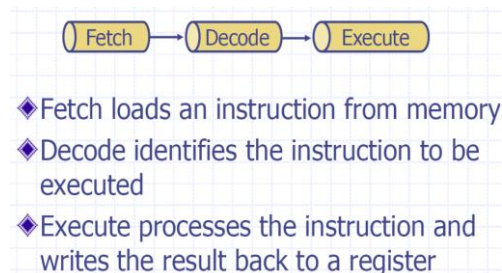
9.

Explain the 3-stage pipelining architecture of ARM7 and the 5-stage pipelining architecture of ARM9 processors with relevant diagrams depicting their pipeline structures.

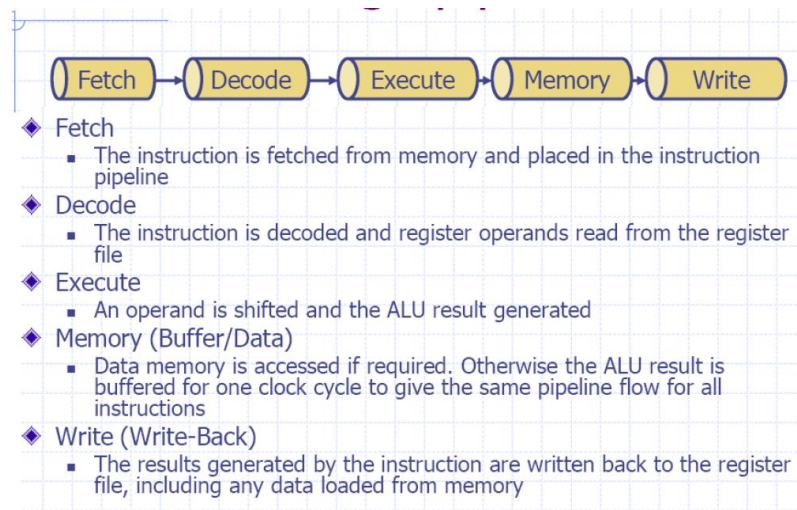
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ANS:

3-stage pipelining architecture of ARM7 processor:



5-stage pipelining architecture of ARM9 processor:



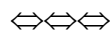
//Explanation for both the architecture should also be included

10. **Provide a concise overview of the data processing instructions available for ARM processors.**

10

ANS:

Overview of the data processing instructions should contain the following topics:
 Data transfer instruction: Various MOV instructions; MOV instruction with Barrel shifter)
 Arithmetic Instructions: ADD, SUB, MUL etc. with and without Barrel shifter
 Logical Instructions: AND, OR, EX-OR etc.
 Compare Instructions: CMP, TEQ, TST etc.



Ca/D/TY

Reg. No: _____



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Final Assessment Test – April 2025

Course: BECE204L - Microprocessors and Microcontrollers

Class NBR(s): 2421 / 2423 / 2424 / 2425 / 2426

Slot: D2+TD2

Max. Marks: 100

Time: Three Hours

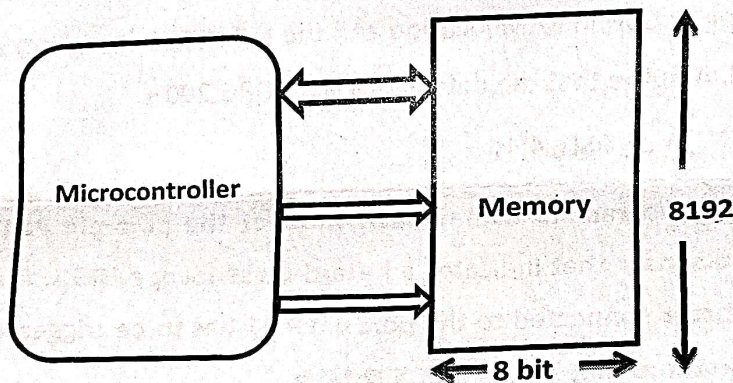
- KEEPING MOBILE PHONE/ANY ELECTRONIC GADGETS, EVEN IN 'OFF' POSITION IS TREATED AS EXAM MALPRACTICE
- DON'T WRITE ANYTHING ON THE QUESTION PAPER

Answer ALL Questions
(10 X 10 = 100 Marks)

1.

(i) Explain why microcontrollers are preferred over microprocessors for small to medium appliance applications.

(ii) Identify the following buses given in the diagram. Also, calculate the size of address bus, data bus and the number of bits to be stored in the given memory.



2.

Explain with suitable diagram the bus interface and execution unit of 8086 microprocessor.

3.

Write an assembly language program to add two 8 bit data. Assume the data as 3EH and 4DH. The first data is in register R0 of bank0 and second data is in register R1 of bank 1. The result should be stored in register R2 of bank 2. Show the status of flags after the addition operation for the above data. Based on the status of each flag(set or reset) store the bit, in bit addressable RAM as follows,

Flags	Bit address(H)
Carry	7F
Aux.carry ✓	6F
Parity ✓	4F

- 4.(a) Write an 8051 assembly program to calculate y which is expressed as $y = x^3 + 9x + 7$ where x is assumed between 0 and 5 and the look-up table for x^3 has to be stored in code space from the location 300H onwards. Store x in register R0 and the final value of y in register R2.

OR

- 4.(b) Write an assembly language program using 8051 microcontroller instructions that finds the number is ODD or EVEN in a given data. Store the ODD data in memory location starting at 60H in RAM location and the EVEN data in memory location starting at 50H. Assume that the data is stored in ORG 200H.

DB 08H, 81H, AFH, 4EH, 68H, 4FH

5. Write an 8051-assembly program to continuously monitor the port pin P1.0 which is connected to a sensor that indicates a hazard by sending a HIGH. If a hazard is detected, a buzzer connected to the port pin P1.1 has to be triggered for 500 microseconds. Assume that XTAL = 11.0592 MHz.

- 6.(a) Write an assembly language program to read port P0 and transmit serially out by giving a copy to P2 do the operations simultaneously at the baud rate of 19200.

OR

- 6.(b) Write an 8051-assembly program that continuously gets 8-bit data from P1 and sends it to P2 while simultaneously creating a square wave of 1KHz on pin P0.1. Use Timer 1 to create the required delay for generating the square wave. Assume that XTAL = 11.0592 MHz.

7. Assume a seven segment LED display is connected in common-cathode configuration at Port1 of 8051 Microcontroller. Develop an ALP to display '0', '2' '4' with delay of 20 msec. Show the interfacing diagram.
8. Assume a switch is connected to P0.7. If SW=0, develop a program to generate a staircase waveform of 10 steps by interfacing DAC0808 with 8051 microcontroller else generate a staircase waveform with 5 steps.
9. Discuss about the dataflow model in an ARM7 with a neat sketch.
10. Explain the following instructions in ARM processor with suitable example.

(a) CMP (b) TEQ (c) B (d) BL (e) MLA

Ca/D/TY